



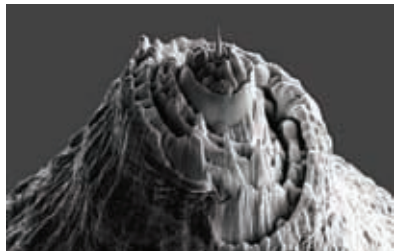
anomaterials are defined as those materials which have structured components with at least one dimension less than 100nm. Much

of nanoscience and many nanotechnologies are concerned with producing new or enhanced “nanomaterials”. These nanomaterials can be constructed by ‘top-down’ as well as by ‘bottom-up’ techniques. Current applications of nanoscale materials include very thin coatings used in electronics and active surfaces.

Based on the number of nano-dimensions, nanomaterials can be:

- ▶ One-dimensional in the nanoscale and extended in the other two dimensions. Examples are layers, such as a thin films or surface coatings.
- ▶ Two-dimensional in the nanoscale and extended in one dimension. Examples include nanowires and nanotubes.
- ▶ Three-dimensional, e.g. Particles like precipitates, colloids and quantum dots.

As we discussed earlier, two principal factors cause the prop-



Tip of Gold imaged by SEM